

Immune System Summary Report

Below is an overview of how your genes relate to immune response. This information is for informational purposes only. Please see the linked articles for details and complete references.

Please wait for the page to load and the genotypes to show up before clicking the print/save button.

[Article: HMGB1](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HMGB1	rs1045411	CC	T	0.23	Increased risk of sepsis;
HMGB1	rs1412125	TC	T	0.48	Increased risk for acquired pneumonia
HMGB1	rs1360485	TT	C	0.3	Increased risk of progression in breast cancer

[Article: Viral Susceptibility](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA-DRB1	rs2187668	CC	T	0.11	Significantly less likely to have SARS coronavirus
CCL2	rs1024611	AG	G	0.28	Increased susceptibility to SARS coronavirus
CD209	rs4804803	AA	G	0.21	Increased susceptibility to malaria; tuberculosis; chronic hepatitis
MBL2	rs1800450	CC	T	0.13	Increased susceptibility to tuberculosis; SARS
OAS1	rs10774671	--	G	0.36	lower risk of West Nile Virus (good)
OAS1	rs2660	AA	G	0.33	Protective against SARS
MX1	rs17000900	--	A	0.07	Protective against SARS
FUT2	rs601338	--	A	0.47	AA: resistant to norovirus and rotavirus
IL1B	rs16944	AG	G	0.63	GG only: less than half the risk of H3N2 flu
IL28	rs8099917	GT	G	0.19	Half the risk for H3N2 flu
TNF	rs361525	GG	A	0.05	AA only: Increased risk for H1N1 flu complications
CD55	rs2564978	--	T	0.24	TT: Increased risk for H1N1 flu
CCR5	rs333	--	D	0.09	DD: resistant to most strains of HIV
TLR3	rs3775291	TC	T	0.28	Decreased risk of hep B; dengue fever and HSV2
IFIH1	rs1990760	CC	T	0.43	Decreased risk for several RNA viruses
CCR5	i3003626	--	D	0.09	DD: resistant to most strains of HIV
intron	rs405103	CC	T	0.14	more likely to have HPV16 persistence >18 months
CSMD1	rs1482207	TT	C	0.17	more likely to have HPV16 persistence >18 months
HLA-DQB1	rs9357152	AA	G	0.26	increased susceptibility to HPV seropositivity (HPV8)

[Article: HIV](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CCR5	i3003626	--	D	0.09	CCR5 delta 32 mutation
CCR5	rs333	--	D	0.09	CCR5 delta 32 mutation

[Article: Response to Vaccines](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA-DQ2	rs2187668	CC	T	0.11	Decreased antibody response to hep. B and measles vaccine
HLA-DPB1	rs3117230	AA	G	0.22	Low antibody response to rubella vaccine
SLAM	rs3796504	GG	T	0.08	4 to 8-fold lower antibody response to measles vaccine
CD46	rs2724384	AA	G	0.2	Lower response to measles vaccine; Increased antibody response to mumps
TLR3	rs7657186	GG	A	0.22	Lower antibody response to MenC vaccine
IL12B	rs3212227	TT	G	0.21	Low response to some flu vaccines
IL-12RB2	rs2201584	--	A	0.16	Lower response to mumps vaccine
IL6	rs1800796	CC	G	0.07	GG: Increased response to some flu vaccine
IL-28B	rs8099917	GT	G	0.19	Twice as likely to produce antibodies for flu vaccine
intergenic	rs10489759	CC	T	0.04	Greatly Decreased response to smallpox vaccine

[Article: Acute Respiratory Distress Syndrome](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
VEGFA	rs3025039	CC	T	0.14	Lower VEGF; Increased risk of mortality in ARDS
ANGPT2	rs2442608	--	C	0.45	CC: Increased ANG2; slightly Increased risk ARDS
ANGPT2	rs2442630	AA	G	0.01	Increased risk of ARDS
ANGPT2	rs2515475	--	T	0.15	Increased risk of ARDS
MBL2	rs1800450	CC	T	0.13	Lower MBL; Increased risk of getting ARDS
MYD88	rs7744	--	G	0.14	Decreased ARDS risk in adults <60
TNF	rs1800629	GG	A	0.15	Increased TNF-alpha; Increased ARDS risk in adults <60
IL17	rs2275913	AA	A	0.33	Decreased ARDS risk (Chinese population); Decreased mortality
NAMPT	rs9770242	--	C	0.23	Slightly Increased risk of ARDS

[Article: Lyme Disease](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TLR1	rs5743618	--	C	0.56	CC only: Increased risk of antibiotic-resistant Lyme arthritis
HLA-DRB1	rs660895	AA	G	0.19	Increased risk of Lyme arthritis
TLR2	rs5743708	--	A	0.02	Reduced risk of Lyme
ANO10	rs41289586	CC	T	0.02	Increased risk of Lyme
ABCB1	rs1128503	AG	G	0.58	AG only: Increased risk of post-treatment Lyme
TLR1	rs4833095	TT	C	0.31	C/C: less of an immune response against B. burgdorferi

[Article: mRNA Side Effects](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA-A	rs2571381	--	T	0.19	HLA-A*03:01 alleles, 2-fold increased risk of severe or extreme side effects from Pfizer COVID-19 vaccine, possibly increased risk for severe or extreme side effects from Moderna
HLA-A	i5018047	--	T	0.19	HLA-A*03:01 alleles, 2-fold increased risk of severe or extreme side effects from Pfizer COVID-19 vaccine, possibly increased risk for severe or extreme side effects from Moderna

[Article: Lyme Susceptibility](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TLR1	rs5743618	--	C	0.56	CC only: Increased risk of antibiotic-resistant Lyme arthritis
TLR1	rs5743708	--	A	0.02	less than half the risk of Lyme
HLA-DRB1	rs660895	AA	G	0.19	increased risk of Lyme arthritis, rheumatoid arthritis.

[Article: Flu](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
IL17A	rs2275913	AA	A	0.33	A/A: ~ half the risk for H3N2 flu
IL1B	rs16944	AG	G	0.63	G/G: half the risk for H3N2 flu
IL28	rs8099917	GT	G	0.19	Half the risk for H3N2 flu
CCR5	i3003626	--	D	0.09	higher risk of flu or increased severity of flu
CD55	rs2564978	--	T	0.24	T/T: increased severity of H1N1 flu infection
MBL2	rs1800450	CC	T	0.13	less likely to have severe H1N1
FCGR2A	rs1801274	AG	A	0.5	increased relative risk of severe pneumonia in H1N1 flu
CCR5	rs333	--	D	0.09	Higher risk of flu or increased severity of flu
FCGR2A	rs9856661	AA	C	0.18	increased relative risk of severe pneumonia in H1N1 flu
TMPRSS2	rs2070788	GG	G	0.45	G/G: higher TMPRSS2 expression, two-fold increased risk of severity with H1N1
TNF	rs361525	GG	A	0.05	increased risk of viral pneumonia with H1N1 flu

[Article: Vitamin D Immune System](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP2R1	rs2060793	AG	A	0.37	Lower vitamin D levels
CYP2R1	rs1562902	CT	T	0.56	Higher vitamin D levels
CYP27B1	rs10877012	--	T	0.29	Lower vitamin D levels
CYP27B1	rs4646536	--	A	0.67	Lower vitamin D levels
CYP27B1	rs28934607	--	A	0	Mutation that causes vitamin D related rickets (rare)
CYP27B1	rs28934605	--	T	0	Mutation that causes vitamin D related rickets (rare)
CYP27B1	rs28934604	--	T	0	Mutation that causes vitamin D related rickets (rare)
GC	rs2282679	TG	G	0.27	Lower vitamin D levels
GC	rs7041	CA	A	0.46	Lower vitamin D levels
GC	rs1155563	TC	C	0.26	Lower vitamin D levels
VDR-Taql	rs731236	AA	A	0.62	AA: Lower vitamin D levels, AG: usually typical vitamin D levels
VDR-Bsml	rs1544410	CC	T	0.38	Lower vitamin D levels
VDR-FokI	rs2228570	--	A	0.38	A/A: slightly longer VDR variant that binds to vitamin D less efficiently; increased relative risk of celiac in people with vitamin D deficiency; A/G: typical
VDR	rs7975232	CC	C	0.44	CC: Half the risk of dengue fever, likely indicates slightly higher vitamin D
GC	rs12512631	--	C	0.35	CC only: slightly higher vitamin D

Article: Cavity Genes

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
DEFB1	rs11362	--	C	0.58	C/C: increased risk of cavities
DEFB1	rs1799946	--	T	0.37	T/T: increased risk of cavities
IL32	rs4786370	--	C	0.41	much more likely to carry the bacteria that causes cavities (S. mutans)
GALK2	rs11635005	--	C	0.8	C/C: Carrying S. mutans significantly increases the risk of cavities, lower GALK2 enzyme activity
AMELX	rs946252	TT	T	0.3	increased risk of cavities
AQP6	rs1996315	--	A	0.55	Decreased risk of cavities
WNT10A	rs121908120	TT	A	0.02	Decreased risk of cavities
C5orf66	rs1122171	--	T	0.57	An average of 1.2 to 2.4 more cavities

Article: Acupuncture

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TRPV1	rs8065080	--	C	0.37	C/C: less TRPV1 activation, acupuncture may be less likely to work
NFKB2	rs1056890	--	G	0.66	G/G: more likely to respond to acupuncture (common genotype)
TCL1A	rs2369049	AA	G	0.14	more likely to respond to acupuncture
COMT	rs4680	AG	A	0.48	lower COMT activity, more likely to respond to acupuncture
COMT	rs6269	AG	G	0.39	A/G: more likely to respond to acupuncture for hot flashes
ANKK1	rs1800497	GA	A	0.2	A/A: Likely less of a response to acupuncture for smoking cessation

Article: Grass Pollen Allergy

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA-DRB4	rs7775228	TC	C	0.13	more likely to be allergic to grass pollen
FLG	rs61816761	GG	A	0.15	(rare mutation) more likely to be allergic to grass pollen
IL2	rs2069762	CA	C	0.28	Increased risk of grass allergy. (Inflammatory cytokine)
IL33	rs928413	AG	G	0.28	Increased risk of hay fever

[Article: Chronic Sinus Infection](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TAS2R38	rs10246939	TT	C	0.46	less likely to have chronic sinus infections
TAS2R38	rs713598	CC	G	0.42	less likely to have chronic sinus infections
NOS1	rs9658281	--	T	0.27	Decreased risk of chronic sinus infections
ALOX15	rs34210653	GG	A	0.01	Significantly reduced risk for nasal polyps and chronic sinus infections
IL1RL1	rs13431828	--	T	0.14	Reduced risk of chronic sinus infections
SERPINA1	rs1243168	GG	A	0.19	A/A: 6-fold increased risk of chronic rhinosinusitis
TNF	rs1800629	GG	A	0.15	Increased TNF-alpha; Increased risk of nasal polyps and sinus infections

[Article: Tamiflu](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CES1	rs71647871	--	T	0.01	60% decreased CES1 enzyme function; decreased conversion of Tamiflu to active medication
CES1	rs121912777	--	T	0.0008	decreased CES1 enzyme function; decreased conversion of Tamiflu to active medication
ABCB1	rs1045642	GG	A	0.49	reduced drug efflux (out of the cell), may need lower dosages of some drugs; Increased risk of adverse events with oseltamivir

[Article: Kawasaki](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
FCGR2A	rs1801274	AG	A	0.5	Increased susceptibility to Kawasaki disease
ITPKC	rs28493229	CG	G	0.87	Increased susceptibility to Kawasaki disease
RAB4B	rs2233152	AG	G	0.86	Increased susceptibility to Kawasaki disease
BLK	rs2736340	TC	T	0.26	Increased susceptibility to Kawasaki disease
CD40	rs1569723	AA	A	0.75	Increased susceptibility to Kawasaki disease, common genotype
INFG	rs1861493	--	G	0.34	Better response to IV immunoglobulin therapy in Kawasaki disease patients

[Article: APOBEC3](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
APOBEC3G	rs8177832	AA	G	0.04	Higher APOBEC3; decreased HIV-1 risk; decreased risk of chronic hepatitis B
APOBEC3F	rs2076101	AG	G	0.48	Lower viral load in HIV with slower progression to AIDS
APOBEC3A	rs12628403	CC	C	0.08	Increased relative risk of breast cancer in some populations
APOBEC3A	rs1014971	CC	T	0.6	increased relative risk of bladder cancer

[Article: IgA Deficiency](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
IFIH1	rs1990760	CC	T	0.43	increased susceptibility to IgA deficiency
IFIH1	rs35667974	TT	C	0.01	protective against IgA deficiency
HLA-DQ2.5	rs2187668	CC	T	0.11	increased risk of IgA deficiency
HLA-DRB1	rs3135388	GG	A	0.11	strongly protective against IgA deficiency

[Article: MRGPRX2 Mast Cell Receptor](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
MRGPRX2	rs10833049	TT	C	0.26	decreased receptor function, decreased mast cell activation
MRGPRX2	rs11024970	--	G	0.05	Change in MRGPRX2 function, responds to pain stimulation (substance P) and slightly increased response to other stimuli

[Article: TLR7](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TLR7	rs179010	TT	T	0.3	increased risk of lupus; lower risk of hepatitis C
TLR7	rs179012	GG	A	0.31	decreased viral load in HIV patients
TLR7	rs179008	AA	T	0.19	increased risk of lupus; decreased risk of severe dengue
TLR7	rs864058	GG	A	0.09	decreased risk of Rift Valley fever virus

[Article: Long Covid](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PEDS1	rs6020298	--	G	0.49	increased long covid susceptibility
NADSYN1	rs12785878	GG	T	0.66	T/T: increased long covid susceptibility
IL10	rs1800896	TT	C	0.45	increased long covid susceptibility
LZTFL1	rs17713054	--	A	0.06	increased long covid susceptibility
SLC15A4	rs11059915	TG	T	0.24	affects long Covid (fatigue) susceptibility
CLOCK	rs62303689	CC	A	0.1	affects long Covid (fatigue) susceptibility
ATP9A	rs77771672	--	G	0.05	affects long Covid (fatigue) susceptibility
GPC5	rs462954	--	G	0.12	affects long Covid (fatigue) susceptibility
INSR	rs8110533	--	T	0.06	affects long Covid (fatigue) susceptibility
FOXP4	rs2496644	CC	A	0.05	increased relative risk of long Covid, increased risk of Covid
LZTFL1	rs10490770	--	C	0.07	increased long covid susceptibility
HLA-DQB1 locus	rs9273363	CC	A	0.27	A/A: no increase in risk of long Covid
IL10RB	rs8178521	--	T	0.21	increased risk of long Covid
ABO	rs8176719	--	D	0.66	D/D: likely to be blood type O; a little less likely to have long Covid in 23andMe GWAS study (also less susceptible to getting Covid)

[Article: Alpha-gal Syndrome](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ABO	rs8176719	--	D	0.63	DD: likely to be blood type O, not protected against alpha-gal syndrome; DI: could be type A or type B
ABO	rs8176746	GG	T	0.08	T/T: likely two type B alleles; less susceptible to alpha-gal; G/T: likely one type B allele (AB or BO)
IL13	rs20541	AA	A	0.2	higher IgE levels; higher risk of allergies in general
IL13	rs1800925	TC	T	0.16	higher IgE levels
STAT6	rs1059513	TT	T	0.9	increased sensitivity to allergens, IgE;
FCER1A	rs2298805	GG	A	0.003	half the risk of chronic urticaria (itchiness); lower serum IgE levels;
FCER1A	rs2251746	TT	C	0.25	lower serum IgE levels
FCER1A	rs2427827	TC	T	0.4	higher serum IgE levels

[Article: Long Covid Study](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HLA-DQB1 locus	rs9273363	CC	C	0.27	slight increase in the relative risk of long Covid
ABO	rs8176719	--	D	0.63	likely to be blood type O, less likely to have long Covid

[Article: COVID-19 & Genetics](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ABO	rs657152	AC	A	0.37	increased risk of COVID-19 severe cases (along with other risk factors such as age and co-morbidities)
ABO	rs10490770	--	C	0.07	increased relative risk for severe COVID-19
F3	rs72729504	CC	T	0.019	~2-fold increased risk of severe COVID-19 outcome
C4BPA	rs45574833	GG	A	0.014	increased risk of hospitalization with COVID-19
Complement system-related	rs12064775	AA	G	0.01	increased risk of hospitalization with COVID-19
TYK2	rs34536443	GG	C	0.03	increased relative risk of severe COVID-19, decreased TYK2
TMEM181	rs117665206	CC	T	0.006	mutation found more frequently in COVID-19 patients who died
ALOXE3	rs147149459	GG	A	0.001	mutation found more frequently in patients who died

[Article: T Cell Exhaustion](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TIM3	rs13170556	TT	C	0.14	increased susceptibility to T cell exhaustion, which increases risk of chronic lung infections
PDCD1	rs2227981	--	A	0.42	Lower PDCD1 expression, theoretically less likely to have T cell exhaustion; reduced cancer risk
PDCD1	rs10204525	--	T	0.15	Likely higher PDCD1 expression, more likely to have T cell exhaustion; increased cancer risk
CTLA4	rs231775	GA	G	0.38	decreased risk of T cell exhaustion in cancer therapy (greater survival rates); increased risk of autoimmune disease
LAG3	rs3782735	AA	G	0.39	increased survival in gastric cancer (likely lower LAG3, less T cell exhaustion)
LRRC8C	rs10493829	--	C	0.41	elevated LRRC8C, improved survival in certain cancers
HLA-DRB1	rs660895	AA	G	0.21	lower risk of T cell exhaustion; higher risk of rheumatoid arthritis (~6-fold) and autoimmune diseases (tag for HLA-DRB1*0401)

[Article: IL-15](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
IL15	rs10833	--	T	.33	T/T: lower IL15; increased liver fibrosis in hepatitis C, but better response to treatment; increased risk of subclinical atherosclerosis; C/T: lower IL15 and increased risk of subclinical atherosclerosis
IL15	rs2857261	--	A	.52	A/A: Increased risk of celiac disease; A/G: no increased risk of celiac
IL15	rs17007695	--	C	.08	increased relative risk of acute lymphoblastic leukemia
IL-15RA	rs2228059	GG	G	.5	increased strength gains in resistance training; decreased relative risk of esophageal or gastric cancer
IL-15RA	rs7097780	AA	G	.32	increased risk of symptomatic osteoarthritis

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